

lecture

October 22, 2025

ranking, correlation clustering, hierarchical clustering

for 2 items, ask oracle for ordering. similarly for 3, ask for which one is outside, for 4, ask for which 2 are grouped.

2 has 1 wrong answer, 3 and 4 have 2 wrong answers

convert the 2 item situation into a directed graph like from 17. $a < b$ implies directed edge from a to b that is positive and a negative from b to a

if m represents the number of constraints, m_s are the number of constraints satisfied, m_v the number violated, and m_a unaffected, we see that

$$W(L, r) = m_s - m_v \quad (1)$$

$$ALG = m_s + \frac{1}{2}(m - m_s - m_v) = \frac{1}{2}(m + m_s - m_v) = \frac{m}{2} + \frac{1}{2}W(L, R) \quad (2)$$

since the weights are bounded, we can increase the weights uniformly (?)

but after increasing, we have to undo the increase which results in a minus total weight term.

for triplets, create a graph that is like a triangle, if bc are together, the edge is -2, ab and ac are +1.

$$W(L, r) = 2m_s - m_v \quad (3)$$

$$ALG = m_s + \frac{1}{3}(m - m_s - m_v) = \frac{1}{3}(m + w(L, r)) \quad (4)$$

for quartets, if ab and cd are together, connect them with -2, then all other configs with +1